

PROBLEM STATEMENT

Triple Composite Chain Rule

Find the derivative with respect to x of the function:

$$y = \sqrt[3]{\log(\sin x)}$$

STEP 1 প্রদত্ত ঘনমূলীয় ফাংশনটিকে সূচকীয় আকারে প্রকাশ করি (Expressing in Power Form)

$$y = [\log(\sin x)]^{1/3}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} ([\log(\sin x)]^{1/3})$$

STEP 3 বাহ্যিক সূচকের জন্য শক্তি বিধি (Power Rule) প্রয়োগ করি (Applying Power Rule to Outer Power)

$$\frac{dy}{dx} = \frac{1}{3} [\log(\sin x)]^{\frac{1}{3}-1} \cdot \frac{d}{dx} [\log(\sin x)] = \frac{1}{3} [\log(\sin x)]^{-2/3} \cdot \frac{d}{dx} [\log(\sin x)]$$

STEP 4 অভ্যন্তরীণ লগারিদমীয় ত্রিকোণমিতিক অংশের অন্তরজ নির্ণয় করি (Differentiating Inner Function via Chain Rule)

$$\frac{d}{dx} [\log(\sin x)] = \frac{1}{\sin x} \cdot \frac{d}{dx} (\sin x) = \frac{1}{\sin x} \cdot \cos x = \cot x$$

STEP 5 মানগুলি একত্রিত করে চূড়ান্ত অন্তরজ পাই (Combining Factors into Final Form)

$$\frac{dy}{dx} = \frac{1}{3 [\log(\sin x)]^{2/3}} \cdot \cot x = \frac{\cot x}{3 [\log(\sin x)]^{2/3}}$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{\cot x}{3 [\log(\sin x)]^{2/3}} \quad \left(\text{or, } \frac{\cot x}{3 \sqrt[3]{\log^2(\sin x)}} \right)$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Composite Hierarchy:** The function has 3 layers: $y = u^{1/3}$ where $u = \log v$ and $v = \sin x$.
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = \left(\frac{1}{3}u^{-2/3}\right) \cdot \left(\frac{1}{v}\right) \cdot (\cos x)$$
- Standard Trigonometric Ratio:** $\frac{\cos x}{\sin x} = \cot x$.
- Power Rule with Fractional Exponent:** $\frac{d}{dx}[u^{1/3}] = \frac{1}{3}u^{-2/3} \frac{du}{dx}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Exponent Misinterpretation:** Confusing $\sqrt[3]{\log(\sin x)} = [\log(\sin x)]^{1/3}$ with $\log(\sin^{1/3} x) = \frac{1}{3} \log(\sin x)$. The cube root applies to the entire logarithm, not the argument!
- Exponent Subtraction Error:** $\frac{1}{3} - 1 = -\frac{2}{3}$. Writing $\frac{2}{3}$ without negative sign creates incorrect numerator placement.
- Missing Inner Derivative:** Forgetting to differentiate $\sin x$ to $\cos x$, erroneously writing $\frac{1}{\sin x}$ alone.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Logarithmic Constraint:** $\sin x > 0 \implies x \in (2k\pi, (2k+1)\pi)$ for $k \in \mathbb{Z}$.
- Denominator Singularity in Derivative:** $\log(\sin x) \neq 0 \implies \sin x \neq 1 \implies x \neq 2k\pi + \frac{\pi}{2}$. At $\sin x = 1$, the tangent line becomes vertical ($\frac{dy}{dx} \rightarrow \infty$).
- Differentiability:** Smooth and differentiable on $(2k\pi, 2k\pi + \frac{\pi}{2}) \cup (2k\pi + \frac{\pi}{2}, (2k+1)\pi)$.

BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural logarithm of both sides:
$$\ln y = \frac{1}{3} \ln(\log(\sin x))$$
- Differentiating implicitly:
$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{3} \cdot \frac{1}{\log(\sin x)} \cdot \frac{\cos x}{\sin x} = \frac{\cot x}{3 \log(\sin x)}$$
- Multiplying by $y = [\log(\sin x)]^{1/3}$:
$$\frac{dy}{dx} = [\log(\sin x)]^{1/3} \cdot \frac{\cot x}{3 \log(\sin x)} = \frac{\cot x}{3[\log(\sin x)]^{2/3}}$$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1–2 (Power Representation & Chain Setup):** 1.0 Mark
- Step 3 (Outer Power Differentiation):** 1.0 Mark
- Step 4 (Inner Logarithmic-Trig Differentiation):** 1.0 Mark
- Step 5 & Final Answer:** 1.0 Mark
- Total Allocation:** 4.0 Marks